

CLASSIFICATION OF SMT MANUFACTURING EQUIPMENT

Placement equipment	Inspection equipment	Soldering equipment	Dispensing equipment
• Low-speed placement • Medium-speed placement • High-speed placement	• Automated optical inspection • X-ray inspection • Solder paste inspection (SPI)	• Reflow soldering • Wave soldering • Selective soldering	• In-line dispenser • Off-line dispenser

Source: Frost & Sullivan

potential in terms of manufacturing and designing capabilities. India's domestic electronics production for the first time increased beyond imports in FY 2016-17, as according to government sources, domestic production stood at US\$ 49.5 billion, higher than the US\$ 43 billion spent on imports.

This seems to be an early signal indicating that the government's Make in India initiative is beginning to make inroads towards improving the state of domestic manufacturing. This, in turn, will drive the demand for manufacturing equipment. According to industry experts, the manufacturing equipment industry in India is worth approximately about half a billion US dollars, including imported and local.

For India, industry experts expect Make in India campaign, key policy initiatives by the government, deep-rooted entrepreneurship spirit, and domestic demand for consumer electronics, telecom equipment, automotive, aeronautics, defence and medical electronics verticals to fuel growth. Moreover, high throughput and high-efficiency manufacturing initiatives, based on Industry 4.0 and Smart Factory—which are being taken up by electronics manufacturers to upgrade product lines—will also augment this pace. SMT will drive the advanced manufacturing equipment market.

Growth drivers

Electronics manufacturing service (EMS) providers need an increasing number of SMT equipment product lines to meet the demand from customers across different industry verticals. In addition, many original design manufacturers (ODMs) have also started manufacturing plants—rather than just designing equipment—to overcome the increasing delays from EMS providers. This will further drive

the growth of the SMT equipment market.

The trend for adopting SMT assembly lines will be bolstered further by increasing demand for higher frequency, highly integrated and smaller devices in consumer electronics, automotive, industrial automation, process control, telecom, the Internet of Things (IoT), computing, storage, aerospace and defence markets.

According to industry experts, major demand-generating sectors are:

- Automotive
- Consumer electronics
- Telecom equipment (including mobile handsets)
- Industrial electronics (including power and energy products)

Demand for SMT devices in the automotive segment has increased due to their ability to meet the needs of critical systems with zero defects. Number of electronic components used in EVs is higher compared to conventional vehicles, since EVs have an electric motor, battery and other complex connections to ensure optimum utilisation of power. Owing to the critical role of electronic components in these EVs, surface-mount devices are used, as these are lighter, occupy less space, are safe and last longer.

The consumer electronics segment is transitioning from manual, analogue systems to digitised, automated control systems. With increased use of smart portable devices and wearables,

there is a heavy size constraint on the PCBs used. This could give a first-mover advantage to a manufacturer who can invest in new SMT equipment that supports miniaturised PCBs or components. Feeder technology, flexible gantries and high-

accuracy optical encoder systems are examples of product innovations as a result of technological advancements. The majority of industry experts indicated that set-top boxes (STBs) is the most demand-generating product segment.

Demand for SMT equipment is also high in the telecom sector (including mobile handset manufacturing) and energy segments.

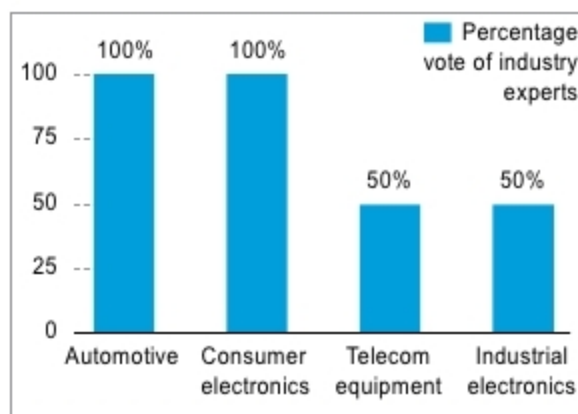
SMT equipment in demand

Placement equipment represents the largest product segment within the SMT equipment market. A steady demand in this segment is attributed to the growth of SMT assembly lines to enable efficient electronics manufacturing practices and innovation related to product miniaturisation and complex circuits. For the same reason, soldering and printing equipment segments are also expected to experience decent demand. Equipment like selective soldering machines, PCB cleaning machines and assembly lines are in demand.

Necessity for production speed enhancement and flexibility of quality assurance procedures has generated demand for advanced SMT inspection systems, including automatic X-ray inspection (AXI) and automatic optical inspection (AOI) systems. AXI equipment is expected to gain from the integration of computer tomography and intuitive programming. Whereas, AOI equipment is expected to become more popular because of the use of machine-vision-based algorithms that enable easy programming, flexibility and enhanced ability to detect faults quickly while minimising false call rates.

Emerging trends in SMT

Industry experts also shared insights on emerging trends that will shape the SMT manufacturing equipment market in India. They believe that



Forecast on major demand-generating application sectors